

Section 1

1.1 Project introduction & problem statement

In recent years, unmanned aerial vehicles (UAVs) have become increasingly important in search and rescue, disaster response, and emergency operations. UAVs can provide aerial information over large areas and access locations that may be difficult or dangerous for humans to reach.

In emergency situations, locating humans quickly is an important task. Traditional search methods may require significant amounts of time and personnel, particularly when the search area is large or difficult to access. Therefore, an automated system capable of detecting and localizing possible humans from UAV data can help improve the efficiency of search operations.

This project proposes an AI-based human detection and localization system capable of identifying possible human locations using UAV camera and sensor data. The system processes information collected by the UAV and estimates the location of detected humans.

The detected information can then be reported to a monitoring system or ground station, allowing operators to view and analyze possible human locations.

1.2 Objectives

1. To develop an AI-based system capable of detecting humans from UAV camera data.
2. To estimate and report the location of detected humans using UAV sensor information.
3. To develop a system capable of processing detection information efficiently.
4. To provide a monitoring interface for displaying UAV and human detection information.
5. To evaluate the performance of the proposed human detection and localization system.

1.3 Scope of the Project

The scope of this project covers the development of a UAV-based human detection and localization system. The system consists of UAV hardware, camera and sensor components, an AI-based detection module, and a monitoring system.

The system focuses on detecting possible humans within the UAV camera field of view and estimating their locations based on available sensor and positioning data.

1.4 Expected Results

The expected result of this project is a prototype system that can detect possible humans from UAV camera data and provide their estimated locations to an operator.

The system is expected to support search and rescue operations by reducing the time required to identify possible human locations and providing useful information for further investigation.